# Aliberiti et al.

| DOMAIN 1: Participants | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 1.1 Were appropriate data sources used, e.g. cohort, RCT or nested case-control study data? | | Yes |
| 1.2 Were all inclusions and exclusions of participants appropriate? | | No |
| **Risk of bias introduced by selection of participants** | **RISK:** (low/ high/ unclear) | High |
| Rationale of bias rating:<br>Only patients aged ≥50 years were included | | |
| **B. Applicability** | | |
| **Concern that the included participants and setting do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 2: Predictors | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 2.1 Were predictors defined and assessed in a similar way for all participants? | | Yes |
| 2.2 Were predictor assessments made without knowledge of outcome data? | | Yes |
| 2.3 Are all predictors available at the time the model is intended to be used? | | Yes |
| **Risk of bias introduced by predictors or their assessment** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| Concern that the definition, assessment or timing of predictors in the model do not match the review question | **CONCERN:** (low/ high/ unclear) | High |
| Rationale of applicability rating:<br>The threshold value of NEWS was not consistent with other studies | | |

| DOMAIN 3: Outcome | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 3.1 Was the outcome determined appropriately? | | Yes |
| 3.2 Was a pre-specified or standard outcome definition used? | | Yes |
| 3.3 Were predictors excluded from the outcome definition? | | Yes |
| 3.4 Was the outcome defined and determined in a similar way for all participants? | | Yes |
| 3.5 Was the outcome determined without knowledge of predictor information? | | Yes |
| 3.6 Was the time interval between predictor assessment and outcome determination appropriate? | | Yes |
| **Risk of bias introduced by the outcome or its determination** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the outcome, its definition, timing or determination do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 4: Analysis | | |
|---|---|---|
| **Risk of Bias** | | |
| 4.1 Were there a reasonable number of participants with the outcome? | | Yes |
| 4.2 Were continuous and categorical predictors handled appropriately? | | Yes |
| 4.3 Were all enrolled participants included in the analysis? | | Yes |
| 4.4 Were participants with missing data handled appropriately? | | Yes |
| 4.5 Was selection of predictors based on univariable analysis avoided? | | Not applicable |
| 4.6 Were complexities in the data (e.g. censoring, competing risks, sampling of controls) accounted for appropriately? | | Yes |
| 4.7 Were relevant model performance measures evaluated appropriately? | | No information |
| 4.8 Were model overfitting and optimism in model performance accounted for? | | Not applicable |
| 4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis? | | Not applicable |
| **Risk of bias introduced by the analysis** | **RISK:** (low/ high/ unclear) | Unclear |
| Rationale of bias rating:<br>Did not report the calibration of the NEWS | | |

# Baker et al.

| DOMAIN 1: Participants | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 1.1 Were appropriate data sources used, e.g. cohort, RCT or nested case-control study data? | | Yes |
| 1.2 Were all inclusions and exclusions of participants appropriate? | | Yes |
| **Risk of bias introduced by selection of participants** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the included participants and setting do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 2: Predictors | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 2.1 Were predictors defined and assessed in a similar way for all participants? | | Yes |
| 2.2 Were predictor assessments made without knowledge of outcome data? | | Yes |
| 2.3 Are all predictors available at the time the model is intended to be used? | | Yes |
| **Risk of bias introduced by predictors or their assessment** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| Concern that the definition, assessment or timing of predictors in the model do not match the review question | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 3: Outcome | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 3.1 Was the outcome determined appropriately? | | Yes |
| 3.2 Was a pre-specified or standard outcome definition used? | | Yes |
| 3.3 Were predictors excluded from the outcome definition? | | Yes |
| 3.4 Was the outcome defined and determined in a similar way for all participants? | | Yes |
| 3.5 Was the outcome determined without knowledge of predictor information? | | Yes |
| 3.6 Was the time interval between predictor assessment and outcome determination appropriate? | | No |
| **Risk of bias introduced by the outcome or its determination** | **RISK:** (low/ high/ unclear) | High |
| Rationale of bias rating:<br>The time interval between predictor assessment and outcome determination was too short | | |
| **B. Applicability** | | |
| **Concern that the outcome, its definition, timing or determination do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| **DOMAIN 4: Analysis** | | |
|---|---|---|
| **Risk of Bias** | | |
| 4.1 Were there a reasonable number of participants with the outcome? | | Yes |
| 4.2 Were continuous and categorical predictors handled appropriately? | | Yes |
| 4.3 Were all enrolled participants included in the analysis? | | Yes |
| 4.4 Were participants with missing data handled appropriately? | | No |
| 4.5 Was selection of predictors based on univariable analysis avoided? | | Not applicable |
| 4.6 Were complexities in the data (e.g. censoring, competing risks, sampling of controls) accounted for appropriately? | | Yes |
| 4.7 Were relevant model performance measures evaluated appropriately? | | No information |
| 4.8 Were model overfitting and optimism in model performance accounted for? | | Not applicable |
| 4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis? | | Not applicable |
| **Risk of bias introduced by the analysis** | **RISK:** (low/ high/ unclear) | High |
| Rationale of bias rating:<br>Patients with missing data were excluded | | |

# Bradley et al.

| DOMAIN 1: Participants | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 1.1 Were appropriate data sources used, e.g. cohort, RCT or nested case-control study data? | | Yes |
| 1.2 Were all inclusions and exclusions of participants appropriate? | | Yes |
| **Risk of bias introduced by selection of participants** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the included participants and setting do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 2: Predictors | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 2.1 Were predictors defined and assessed in a similar way for all participants? | | Yes |
| 2.2 Were predictor assessments made without knowledge of outcome data? | | Yes |
| 2.3 Are all predictors available at the time the model is intended to be used? | | Yes |
| **Risk of bias introduced by predictors or their assessment** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| Concern that the definition, assessment or timing of predictors in the model do not match the review question | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 3: Outcome | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 3.1 Was the outcome determined appropriately? | | Yes |
| 3.2 Was a pre-specified or standard outcome definition used? | | Yes |
| 3.3 Were predictors excluded from the outcome definition? | | Yes |
| 3.4 Was the outcome defined and determined in a similar way for all participants? | | Yes |
| 3.5 Was the outcome determined without knowledge of predictor information? | | Yes |
| 3.6 Was the time interval between predictor assessment and outcome determination appropriate? | | No information |
| **Risk of bias introduced by the outcome or its determination** | **RISK:** (low/ high/ unclear) | Unclear |
| Rationale of bias rating:<br>The time interval between predictor assessment and outcome determination was not specified | | |
| **B. Applicability** | | |
| **Concern that the outcome, its definition, timing or determination do not match the review question** | **CONCERN:** (low/ high/ unclear) | Unclear |
| Rationale of applicability rating:<br>The time interval between predictor assessment and outcome determination was not specified | | |

| **DOMAIN 4: Analysis** | | |
|---|---|---|
| **Risk of Bias** | | |
| 4.1 Were there a reasonable number of participants with the outcome? | | Yes |
| 4.2 Were continuous and categorical predictors handled appropriately? | | Yes |
| 4.3 Were all enrolled participants included in the analysis? | | Yes |
| 4.4 Were participants with missing data handled appropriately? | | No |
| 4.5 Was selection of predictors based on univariable analysis avoided? | | Not applicable |
| 4.6 Were complexities in the data (e.g. censoring, competing risks, sampling of controls) accounted for appropriately? | | Yes |
| 4.7 Were relevant model performance measures evaluated appropriately? | | Yes |
| 4.8 Were model overfitting and optimism in model performance accounted for? | | Not applicable |
| 4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis? | | Not applicable |
| **Risk of bias introduced by the analysis** | **RISK:** (low/ high/ unclear) | High |
| Rationale of bias rating:<br>Patients with missing data were excluded | | |

# Covino et al.

| DOMAIN 1: Participants | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 1.1 Were appropriate data sources used, e.g. cohort, RCT or nested case-control study data? | | Yes |
| 1.2 Were all inclusions and exclusions of participants appropriate? | | Yes |
| **Risk of bias introduced by selection of participants** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the included participants and setting do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 2: Predictors | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 2.1 Were predictors defined and assessed in a similar way for all participants? | | Yes |
| 2.2 Were predictor assessments made without knowledge of outcome data? | | Yes |
| 2.3 Are all predictors available at the time the model is intended to be used? | | Yes |
| **Risk of bias introduced by predictors or their assessment** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| Concern that the definition, assessment or timing of predictors in the model do not match the review question | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 3: Outcome | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 3.1 Was the outcome determined appropriately? | | Yes |
| 3.2 Was a pre-specified or standard outcome definition used? | | Yes |
| 3.3 Were predictors excluded from the outcome definition? | | Yes |
| 3.4 Was the outcome defined and determined in a similar way for all participants? | | Yes |
| 3.5 Was the outcome determined without knowledge of predictor information? | | Yes |
| 3.6 Was the time interval between predictor assessment and outcome determination appropriate? | | Yes |
| **Risk of bias introduced by the outcome or its determination** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the outcome, its definition, timing or determination do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| **DOMAIN 4: Analysis** | | |
|---|---|---|
| **Risk of Bias** | | |
| 4.1 Were there a reasonable number of participants with the outcome? | | Yes |
| 4.2 Were continuous and categorical predictors handled appropriately? | | Yes |
| 4.3 Were all enrolled participants included in the analysis? | | Yes |
| 4.4 Were participants with missing data handled appropriately? | | No |
| 4.5 Was selection of predictors based on univariable analysis avoided? | | Not applicable |
| 4.6 Were complexities in the data (e.g. censoring, competing risks, sampling of controls) accounted for appropriately? | | Yes |
| 4.7 Were relevant model performance measures evaluated appropriately? | | No information |
| 4.8 Were model overfitting and optimism in model performance accounted for? | | Not applicable |
| 4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis? | | Not applicable |
| **Risk of bias introduced by the analysis** | **RISK:** (low/ high/ unclear) | High |
| Rationale of bias rating:<br>Patients with missing data were excluded | | |

# De Socio et al.

| DOMAIN 1: Participants | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 1.1 Were appropriate data sources used, e.g. cohort, RCT or nested case-control study data? | | Yes |
| 1.2 Were all inclusions and exclusions of participants appropriate? | | Yes |
| **Risk of bias introduced by selection of participants** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the included participants and setting do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 2: Predictors | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 2.1 Were predictors defined and assessed in a similar way for all participants? | | Yes |
| 2.2 Were predictor assessments made without knowledge of outcome data? | | Yes |
| 2.3 Are all predictors available at the time the model is intended to be used? | | Yes |
| **Risk of bias introduced by predictors or their assessment** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| Concern that the definition, assessment or timing of predictors in the model do not match the review question | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 3: Outcome | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 3.1 Was the outcome determined appropriately? | | Yes |
| 3.2 Was a pre-specified or standard outcome definition used? | | Yes |
| 3.3 Were predictors excluded from the outcome definition? | | Yes |
| 3.4 Was the outcome defined and determined in a similar way for all participants? | | Yes |
| 3.5 Was the outcome determined without knowledge of predictor information? | | Yes |
| 3.6 Was the time interval between predictor assessment and outcome determination appropriate? | | Yes |
| **Risk of bias introduced by the outcome or its determination** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the outcome, its definition, timing or determination do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| **DOMAIN 4: Analysis** | | |
|---|---|---|
| **Risk of Bias** | | |
| 4.1 Were there a reasonable number of participants with the outcome? | | Yes |
| 4.2 Were continuous and categorical predictors handled appropriately? | | Yes |
| 4.3 Were all enrolled participants included in the analysis? | | Yes |
| 4.4 Were participants with missing data handled appropriately? | | No |
| 4.5 Was selection of predictors based on univariable analysis avoided? | | Not applicable |
| 4.6 Were complexities in the data (e.g. censoring, competing risks, sampling of controls) accounted for appropriately? | | Yes |
| 4.7 Were relevant model performance measures evaluated appropriately? | | No information |
| 4.8 Were model overfitting and optimism in model performance accounted for? | | Not applicable |
| 4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis? | | Not applicable |
| **Risk of bias introduced by the analysis** | **RISK:** (low/ high/ unclear) | High |
| Rationale of bias rating:<br>Patients with missing data were excluded | | |

# Fan et al.

| DOMAIN 1: Participants | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 1.1 Were appropriate data sources used, e.g. cohort, RCT or nested case-control study data? | | Yes |
| 1.2 Were all inclusions and exclusions of participants appropriate? | | No information |
| **Risk of bias introduced by selection of participants** | **RISK:** (low/ high/ unclear) | Unclear |
| Rationale of bias rating:<br>The exclusions of participants were not specified | | |
| **B. Applicability** | | |
| **Concern that the included participants and setting do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 2: Predictors | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 2.1 Were predictors defined and assessed in a similar way for all participants? | | Yes |
| 2.2 Were predictor assessments made without knowledge of outcome data? | | Yes |
| 2.3 Are all predictors available at the time the model is intended to be used? | | Yes |
| **Risk of bias introduced by predictors or their assessment** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| Concern that the definition, assessment or timing of predictors in the model do not match the review question | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 3: Outcome | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 3.1 Was the outcome determined appropriately? | | Yes |
| 3.2 Was a pre-specified or standard outcome definition used? | | Yes |
| 3.3 Were predictors excluded from the outcome definition? | | Yes |
| 3.4 Was the outcome defined and determined in a similar way for all participants? | | Yes |
| 3.5 Was the outcome determined without knowledge of predictor information? | | Yes |
| 3.6 Was the time interval between predictor assessment and outcome determination appropriate? | | Yes |
| **Risk of bias introduced by the outcome or its determination** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the outcome, its definition, timing or determination do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| **DOMAIN 4: Analysis** | | |
|---|---|---|
| **Risk of Bias** | | |
| 4.1 Were there a reasonable number of participants with the outcome? | | Yes |
| 4.2 Were continuous and categorical predictors handled appropriately? | | Yes |
| 4.3 Were all enrolled participants included in the analysis? | | Yes |
| 4.4 Were participants with missing data handled appropriately? | | No |
| 4.5 Was selection of predictors based on univariable analysis avoided? | | Not applicable |
| 4.6 Were complexities in the data (e.g. censoring, competing risks, sampling of controls) accounted for appropriately? | | Yes |
| 4.7 Were relevant model performance measures evaluated appropriately? | | No information |
| 4.8 Were model overfitting and optimism in model performance accounted for? | | Not applicable |
| 4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis? | | Not applicable |
| **Risk of bias introduced by the analysis** | **RISK:** (low/ high/ unclear) | High |
| Rationale of bias rating:<br>Patients with missing data were excluded | | |

# Gidari et al.

| DOMAIN 1: Participants | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 1.1 Were appropriate data sources used, e.g. cohort, RCT or nested case-control study data? | | Yes |
| 1.2 Were all inclusions and exclusions of participants appropriate? | | Yes |
| **Risk of bias introduced by selection of participants** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the included participants and setting do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 2: Predictors | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 2.1 Were predictors defined and assessed in a similar way for all participants? | | Yes |
| 2.2 Were predictor assessments made without knowledge of outcome data? | | Yes |
| 2.3 Are all predictors available at the time the model is intended to be used? | | Yes |
| **Risk of bias introduced by predictors or their assessment** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| Concern that the definition, assessment or timing of predictors in the model do not match the review question | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 3: Outcome | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 3.1 Was the outcome determined appropriately? | | Yes |
| 3.2 Was a pre-specified or standard outcome definition used? | | Yes |
| 3.3 Were predictors excluded from the outcome definition? | | Yes |
| 3.4 Was the outcome defined and determined in a similar way for all participants? | | Yes |
| 3.5 Was the outcome determined without knowledge of predictor information? | | Yes |
| 3.6 Was the time interval between predictor assessment and outcome determination appropriate? | | Yes |
| **Risk of bias introduced by the outcome or its determination** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the outcome, its definition, timing or determination do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| **DOMAIN 4: Analysis** | | |
|---|---|---|
| **Risk of Bias** | | |
| 4.1 Were there a reasonable number of participants with the outcome? | | No |
| 4.2 Were continuous and categorical predictors handled appropriately? | | Yes |
| 4.3 Were all enrolled participants included in the analysis? | | Yes |
| 4.4 Were participants with missing data handled appropriately? | | No information |
| 4.5 Was selection of predictors based on univariable analysis avoided? | | Not applicable |
| 4.6 Were complexities in the data (e.g. censoring, competing risks, sampling of controls) accounted for appropriately? | | Yes |
| 4.7 Were relevant model performance measures evaluated appropriately? | | No information |
| 4.8 Were model overfitting and optimism in model performance accounted for? | | Not applicable |
| 4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis? | | Not applicable |
| **Risk of bias introduced by the analysis** | **RISK:** (low/ high/ unclear) | High |
| Rationale of bias rating:<br>Sample size was less than 100 | | |

# Holten et al.

| DOMAIN 1: Participants | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 1.1 Were appropriate data sources used, e.g. cohort, RCT or nested case-control study data? | | Yes |
| 1.2 Were all inclusions and exclusions of participants appropriate? | | Yes |
| **Risk of bias introduced by selection of participants** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the included participants and setting do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 2: Predictors | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 2.1 Were predictors defined and assessed in a similar way for all participants? | | Yes |
| 2.2 Were predictor assessments made without knowledge of outcome data? | | Yes |
| 2.3 Are all predictors available at the time the model is intended to be used? | | Yes |
| **Risk of bias introduced by predictors or their assessment** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| Concern that the definition, assessment or timing of predictors in the model do not match the review question | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 3: Outcome | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 3.1 Was the outcome determined appropriately? | | Yes |
| 3.2 Was a pre-specified or standard outcome definition used? | | Yes |
| 3.3 Were predictors excluded from the outcome definition? | | Yes |
| 3.4 Was the outcome defined and determined in a similar way for all participants? | | Yes |
| 3.5 Was the outcome determined without knowledge of predictor information? | | Yes |
| 3.6 Was the time interval between predictor assessment and outcome determination appropriate? | | Yes |
| **Risk of bias introduced by the outcome or its determination** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the outcome, its definition, timing or determination do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| **DOMAIN 4: Analysis** | | |
|---|---|---|
| **Risk of Bias** | | |
| 4.1 Were there a reasonable number of participants with the outcome? | | Yes |
| 4.2 Were continuous and categorical predictors handled appropriately? | | Yes |
| 4.3 Were all enrolled participants included in the analysis? | | Yes |
| 4.4 Were participants with missing data handled appropriately? | | No information |
| 4.5 Was selection of predictors based on univariable analysis avoided? | | Not applicable |
| 4.6 Were complexities in the data (e.g. censoring, competing risks, sampling of controls) accounted for appropriately? | | Yes |
| 4.7 Were relevant model performance measures evaluated appropriately? | | No information |
| 4.8 Were model overfitting and optimism in model performance accounted for? | | Not applicable |
| 4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis? | | Not applicable |
| **Risk of bias introduced by the analysis** | **RISK:** (low/ high/ unclear) | Unclear |
| Rationale of bias rating:<br>Did not report the missing data and the calibration of NEWS | | |

# Ihle-Hansen et al.

| DOMAIN 1: Participants | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 1.1 Were appropriate data sources used, e.g. cohort, RCT or nested case-control study data? | | Yes |
| 1.2 Were all inclusions and exclusions of participants appropriate? | | Yes |
| **Risk of bias introduced by selection of participants** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the included participants and setting do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 2: Predictors | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 2.1 Were predictors defined and assessed in a similar way for all participants? | | Yes |
| 2.2 Were predictor assessments made without knowledge of outcome data? | | Yes |
| 2.3 Are all predictors available at the time the model is intended to be used? | | Yes |
| **Risk of bias introduced by predictors or their assessment** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| Concern that the definition, assessment or timing of predictors in the model do not match the review question | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 3: Outcome | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 3.1 Was the outcome determined appropriately? | | Yes |
| 3.2 Was a pre-specified or standard outcome definition used? | | Yes |
| 3.3 Were predictors excluded from the outcome definition? | | Yes |
| 3.4 Was the outcome defined and determined in a similar way for all participants? | | Yes |
| 3.5 Was the outcome determined without knowledge of predictor information? | | Yes |
| 3.6 Was the time interval between predictor assessment and outcome determination appropriate? | | No information |
| **Risk of bias introduced by the outcome or its determination** | **RISK:** (low/ high/ unclear) | Unclear |
| Rationale of bias rating:<br>The time interval between predictor assessment and outcome determination was not specified | | |
| **B. Applicability** | | |
| **Concern that the outcome, its definition, timing or determination do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| **DOMAIN 4: Analysis** | | |
|---|---|---|
| **Risk of Bias** | | |
| 4.1 Were there a reasonable number of participants with the outcome? | | No |
| 4.2 Were continuous and categorical predictors handled appropriately? | | Yes |
| 4.3 Were all enrolled participants included in the analysis? | | Yes |
| 4.4 Were participants with missing data handled appropriately? | | No information |
| 4.5 Was selection of predictors based on univariable analysis avoided? | | Not applicable |
| 4.6 Were complexities in the data (e.g. censoring, competing risks, sampling of controls) accounted for appropriately? | | Yes |
| 4.7 Were relevant model performance measures evaluated appropriately? | | No information |
| 4.8 Were model overfitting and optimism in model performance accounted for? | | Not applicable |
| 4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis? | | Not applicable |
| **Risk of bias introduced by the analysis** | **RISK:** (low/ high/ unclear) | High |
| Rationale of bias rating:<br>Sample size was less than 100 | | |

# Jang et al.

| DOMAIN 1: Participants | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 1.1 Were appropriate data sources used, e.g. cohort, RCT or nested case-control study data? | | Yes |
| 1.2 Were all inclusions and exclusions of participants appropriate? | | Yes |
| **Risk of bias introduced by selection of participants** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the included participants and setting do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 2: Predictors | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 2.1 Were predictors defined and assessed in a similar way for all participants? | | Yes |
| 2.2 Were predictor assessments made without knowledge of outcome data? | | Yes |
| 2.3 Are all predictors available at the time the model is intended to be used? | | Yes |
| **Risk of bias introduced by predictors or their assessment** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| Concern that the definition, assessment or timing of predictors in the model do not match the review question | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 3: Outcome | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 3.1 Was the outcome determined appropriately? | | Yes |
| 3.2 Was a pre-specified or standard outcome definition used? | | Yes |
| 3.3 Were predictors excluded from the outcome definition? | | Yes |
| 3.4 Was the outcome defined and determined in a similar way for all participants? | | Yes |
| 3.5 Was the outcome determined without knowledge of predictor information? | | Yes |
| 3.6 Was the time interval between predictor assessment and outcome determination appropriate? | | No information |
| **Risk of bias introduced by the outcome or its determination** | **RISK:** (low/ high/ unclear) | Unclear |
| Rationale of bias rating:<br>The time interval between predictor assessment and outcome determination was not specified | | |
| **B. Applicability** | | |
| **Concern that the outcome, its definition, timing or determination do not match the review question** | **CONCERN:** (low/ high/ unclear) | Unclear |
| Rationale of applicability rating:<br>The time interval between predictor assessment and outcome determination was not specified | | |

| **DOMAIN 4: Analysis** | | |
|---|---|---|
| **Risk of Bias** | | |
| 4.1 Were there a reasonable number of participants with the outcome? | | Yes |
| 4.2 Were continuous and categorical predictors handled appropriately? | | Yes |
| 4.3 Were all enrolled participants included in the analysis? | | Yes |
| 4.4 Were participants with missing data handled appropriately? | | No information |
| 4.5 Was selection of predictors based on univariable analysis avoided? | | Not applicable |
| 4.6 Were complexities in the data (e.g. censoring, competing risks, sampling of controls) accounted for appropriately? | | Yes |
| 4.7 Were relevant model performance measures evaluated appropriately? | | No information |
| 4.8 Were model overfitting and optimism in model performance accounted for? | | Not applicable |
| 4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis? | | Not applicable |
| **Risk of bias introduced by the analysis** | **RISK:** (low/ high/ unclear) | High |
| Rationale of bias rating:<br>Did not report the missing data and the calibration of NEWS | | |

# Liu et al.

| DOMAIN 1: Participants | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 1.1 Were appropriate data sources used, e.g. cohort, RCT or nested case-control study data? | | Yes |
| 1.2 Were all inclusions and exclusions of participants appropriate? | | Yes |
| **Risk of bias introduced by selection of participants** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the included participants and setting do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 2: Predictors | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 2.1 Were predictors defined and assessed in a similar way for all participants? | | Yes |
| 2.2 Were predictor assessments made without knowledge of outcome data? | | Yes |
| 2.3 Are all predictors available at the time the model is intended to be used? | | Yes |
| **Risk of bias introduced by predictors or their assessment** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| Concern that the definition, assessment or timing of predictors in the model do not match the review question | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 3: Outcome | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 3.1 Was the outcome determined appropriately? | | Yes |
| 3.2 Was a pre-specified or standard outcome definition used? | | Yes |
| 3.3 Were predictors excluded from the outcome definition? | | Yes |
| 3.4 Was the outcome defined and determined in a similar way for all participants? | | Yes |
| 3.5 Was the outcome determined without knowledge of predictor information? | | Yes |
| 3.6 Was the time interval between predictor assessment and outcome determination appropriate? | | Yes |
| **Risk of bias introduced by the outcome or its determination** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the outcome, its definition, timing or determination do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| **DOMAIN 4: Analysis** | | |
|---|---|---|
| **Risk of Bias** | | |
| 4.1 Were there a reasonable number of participants with the outcome? | | Yes |
| 4.2 Were continuous and categorical predictors handled appropriately? | | Yes |
| 4.3 Were all enrolled participants included in the analysis? | | Yes |
| 4.4 Were participants with missing data handled appropriately? | | Yes |
| 4.5 Was selection of predictors based on univariable analysis avoided? | | Not applicable |
| 4.6 Were complexities in the data (e.g. censoring, competing risks, sampling of controls) accounted for appropriately? | | Yes |
| 4.7 Were relevant model performance measures evaluated appropriately? | | Yes |
| 4.8 Were model overfitting and optimism in model performance accounted for? | | Not applicable |
| 4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis? | | Not applicable |
| **Risk of bias introduced by the analysis** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |

# Maguire et al.

| DOMAIN 1: Participants | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 1.1 Were appropriate data sources used, e.g. cohort, RCT or nested case-control study data? | | Yes |
| 1.2 Were all inclusions and exclusions of participants appropriate? | | Yes |
| **Risk of bias introduced by selection of participants** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the included participants and setting do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 2: Predictors | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 2.1 Were predictors defined and assessed in a similar way for all participants? | | Yes |
| 2.2 Were predictor assessments made without knowledge of outcome data? | | Yes |
| 2.3 Are all predictors available at the time the model is intended to be used? | | Yes |
| **Risk of bias introduced by predictors or their assessment** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| Concern that the definition, assessment or timing of predictors in the model do not match the review question | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 3: Outcome | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 3.1 Was the outcome determined appropriately? | | Yes |
| 3.2 Was a pre-specified or standard outcome definition used? | | Yes |
| 3.3 Were predictors excluded from the outcome definition? | | Yes |
| 3.4 Was the outcome defined and determined in a similar way for all participants? | | Yes |
| 3.5 Was the outcome determined without knowledge of predictor information? | | Yes |
| 3.6 Was the time interval between predictor assessment and outcome determination appropriate? | | Yes |
| **Risk of bias introduced by the outcome or its determination** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the outcome, its definition, timing or determination do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| **DOMAIN 4: Analysis** | | |
|---|---|---|
| **Risk of Bias** | | |
| 4.1 Were there a reasonable number of participants with the outcome? | | Yes |
| 4.2 Were continuous and categorical predictors handled appropriately? | | Yes |
| 4.3 Were all enrolled participants included in the analysis? | | Yes |
| 4.4 Were participants with missing data handled appropriately? | | No |
| 4.5 Was selection of predictors based on univariable analysis avoided? | | Not applicable |
| 4.6 Were complexities in the data (e.g. censoring, competing risks, sampling of controls) accounted for appropriately? | | Yes |
| 4.7 Were relevant model performance measures evaluated appropriately? | | No information |
| 4.8 Were model overfitting and optimism in model performance accounted for? | | Not applicable |
| 4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis? | | Not applicable |
| **Risk of bias introduced by the analysis** | **RISK:** (low/ high/ unclear) | High |
| Rationale of bias rating:<br>Patients with missing data were excluded | | |

# Martín-Rodríguez et al.

| DOMAIN 1: Participants | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 1.1 Were appropriate data sources used, e.g. cohort, RCT or nested case-control study data? | | Yes |
| 1.2 Were all inclusions and exclusions of participants appropriate? | | Yes |
| **Risk of bias introduced by selection of participants** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the included participants and setting do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 2: Predictors | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 2.1 Were predictors defined and assessed in a similar way for all participants? | | Yes |
| 2.2 Were predictor assessments made without knowledge of outcome data? | | Yes |
| 2.3 Are all predictors available at the time the model is intended to be used? | | Yes |
| **Risk of bias introduced by predictors or their assessment** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| Concern that the definition, assessment or timing of predictors in the model do not match the review question | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 3: Outcome | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 3.1 Was the outcome determined appropriately? | | Yes |
| 3.2 Was a pre-specified or standard outcome definition used? | | Yes |
| 3.3 Were predictors excluded from the outcome definition? | | Yes |
| 3.4 Was the outcome defined and determined in a similar way for all participants? | | Yes |
| 3.5 Was the outcome determined without knowledge of predictor information? | | Yes |
| 3.6 Was the time interval between predictor assessment and outcome determination appropriate? | | Yes |
| **Risk of bias introduced by the outcome or its determination** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the outcome, its definition, timing or determination do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| **DOMAIN 4: Analysis** | | |
|---|---|---|
| **Risk of Bias** | | |
| 4.1 Were there a reasonable number of participants with the outcome? | | Yes |
| 4.2 Were continuous and categorical predictors handled appropriately? | | Yes |
| 4.3 Were all enrolled participants included in the analysis? | | Yes |
| 4.4 Were participants with missing data handled appropriately? | | Yes |
| 4.5 Was selection of predictors based on univariable analysis avoided? | | Not applicable |
| 4.6 Were complexities in the data (e.g. censoring, competing risks, sampling of controls) accounted for appropriately? | | Yes |
| 4.7 Were relevant model performance measures evaluated appropriately? | | Yes |
| 4.8 Were model overfitting and optimism in model performance accounted for? | | Not applicable |
| 4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis? | | Not applicable |
| **Risk of bias introduced by the analysis** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Missing values were replaced using the mode of the variable | | |

# Myrstad et al.

| DOMAIN 1: Participants | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 1.1 Were appropriate data sources used, e.g. cohort, RCT or nested case-control study data? | | Yes |
| 1.2 Were all inclusions and exclusions of participants appropriate? | | Yes |
| **Risk of bias introduced by selection of participants** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the included participants and setting do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 2: Predictors | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 2.1 Were predictors defined and assessed in a similar way for all participants? | | Yes |
| 2.2 Were predictor assessments made without knowledge of outcome data? | | Yes |
| 2.3 Are all predictors available at the time the model is intended to be used? | | Yes |
| **Risk of bias introduced by predictors or their assessment** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| Concern that the definition, assessment or timing of predictors in the model do not match the review question | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 3: Outcome | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 3.1 Was the outcome determined appropriately? | | Yes |
| 3.2 Was a pre-specified or standard outcome definition used? | | Yes |
| 3.3 Were predictors excluded from the outcome definition? | | Yes |
| 3.4 Was the outcome defined and determined in a similar way for all participants? | | Yes |
| 3.5 Was the outcome determined without knowledge of predictor information? | | Yes |
| 3.6 Was the time interval between predictor assessment and outcome determination appropriate? | | Yes |
| **Risk of bias introduced by the outcome or its determination** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the outcome, its definition, timing or determination do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| **DOMAIN 4: Analysis** | | |
|---|---|---|
| **Risk of Bias** | | |
| 4.1 Were there a reasonable number of participants with the outcome? | | No |
| 4.2 Were continuous and categorical predictors handled appropriately? | | Yes |
| 4.3 Were all enrolled participants included in the analysis? | | Yes |
| 4.4 Were participants with missing data handled appropriately? | | Yes |
| 4.5 Was selection of predictors based on univariable analysis avoided? | | Not applicable |
| 4.6 Were complexities in the data (e.g. censoring, competing risks, sampling of controls) accounted for appropriately? | | Yes |
| 4.7 Were relevant model performance measures evaluated appropriately? | | No information |
| 4.8 Were model overfitting and optimism in model performance accounted for? | | Not applicable |
| 4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis? | | Not applicable |
| **Risk of bias introduced by the analysis** | **RISK:** (low/ high/ unclear) | High |
| Rationale of bias rating:<br>Sample size was less than 100 | | |

# Pokeerbux et al.

| DOMAIN 1: Participants | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 1.1 Were appropriate data sources used, e.g. cohort, RCT or nested case-control study data? | | Yes |
| 1.2 Were all inclusions and exclusions of participants appropriate? | | Yes |
| **Risk of bias introduced by selection of participants** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the included participants and setting do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 2: Predictors | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 2.1 Were predictors defined and assessed in a similar way for all participants? | | Yes |
| 2.2 Were predictor assessments made without knowledge of outcome data? | | Yes |
| 2.3 Are all predictors available at the time the model is intended to be used? | | Yes |
| **Risk of bias introduced by predictors or their assessment** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| Concern that the definition, assessment or timing of predictors in the model do not match the review question | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 3: Outcome | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 3.1 Was the outcome determined appropriately? | | Yes |
| 3.2 Was a pre-specified or standard outcome definition used? | | Yes |
| 3.3 Were predictors excluded from the outcome definition? | | Yes |
| 3.4 Was the outcome defined and determined in a similar way for all participants? | | Yes |
| 3.5 Was the outcome determined without knowledge of predictor information? | | Yes |
| 3.6 Was the time interval between predictor assessment and outcome determination appropriate? | | Yes |
| **Risk of bias introduced by the outcome or its determination** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the outcome, its definition, timing or determination do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| **DOMAIN 4: Analysis** | | |
|---|---|---|
| **Risk of Bias** | | |
| 4.1 Were there a reasonable number of participants with the outcome? | | Yes |
| 4.2 Were continuous and categorical predictors handled appropriately? | | Yes |
| 4.3 Were all enrolled participants included in the analysis? | | Yes |
| 4.4 Were participants with missing data handled appropriately? | | No information |
| 4.5 Was selection of predictors based on univariable analysis avoided? | | Not applicable |
| 4.6 Were complexities in the data (e.g. censoring, competing risks, sampling of controls) accounted for appropriately? | | Yes |
| 4.7 Were relevant model performance measures evaluated appropriately? | | No information |
| 4.8 Were model overfitting and optimism in model performance accounted for? | | Not applicable |
| 4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis? | | Not applicable |
| **Risk of bias introduced by the analysis** | **RISK:** (low/ high/ unclear) | Unclear |
| Rationale of bias rating:<br>Did not report the missing data and the calibration of NEWS | | |

# Prower et al.

| DOMAIN 1: Participants | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 1.1 Were appropriate data sources used, e.g. cohort, RCT or nested case-control study data? | | Yes |
| 1.2 Were all inclusions and exclusions of participants appropriate? | | Yes |
| **Risk of bias introduced by selection of participants** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the included participants and setting do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 2: Predictors | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 2.1 Were predictors defined and assessed in a similar way for all participants? | | Yes |
| 2.2 Were predictor assessments made without knowledge of outcome data? | | Yes |
| 2.3 Are all predictors available at the time the model is intended to be used? | | Yes |
| **Risk of bias introduced by predictors or their assessment** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| Concern that the definition, assessment or timing of predictors in the model do not match the review question | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 3: Outcome | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 3.1 Was the outcome determined appropriately? | | Yes |
| 3.2 Was a pre-specified or standard outcome definition used? | | Yes |
| 3.3 Were predictors excluded from the outcome definition? | | Yes |
| 3.4 Was the outcome defined and determined in a similar way for all participants? | | Yes |
| 3.5 Was the outcome determined without knowledge of predictor information? | | Yes |
| 3.6 Was the time interval between predictor assessment and outcome determination appropriate? | | Yes |
| **Risk of bias introduced by the outcome or its determination** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the outcome, its definition, timing or determination do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| **DOMAIN 4: Analysis** | | |
|---|---|---|
| **Risk of Bias** | | |
| 4.1 Were there a reasonable number of participants with the outcome? | | Yes |
| 4.2 Were continuous and categorical predictors handled appropriately? | | Yes |
| 4.3 Were all enrolled participants included in the analysis? | | Yes |
| 4.4 Were participants with missing data handled appropriately? | | No information |
| 4.5 Was selection of predictors based on univariable analysis avoided? | | Not applicable |
| 4.6 Were complexities in the data (e.g. censoring, competing risks, sampling of controls) accounted for appropriately? | | Yes |
| 4.7 Were relevant model performance measures evaluated appropriately? | | No information |
| 4.8 Were model overfitting and optimism in model performance accounted for? | | Not applicable |
| 4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis? | | Not applicable |
| **Risk of bias introduced by the analysis** | **RISK:** (low/ high/ unclear) | Unclear |
| Rationale of bias rating:<br>Did not report the missing data and the calibration of NEWS | | |

# Richardson et al.

| DOMAIN 1: Participants | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 1.1 Were appropriate data sources used, e.g. cohort, RCT or nested case-control study data? | | Yes |
| 1.2 Were all inclusions and exclusions of participants appropriate? | | Yes |
| **Risk of bias introduced by selection of participants** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the included participants and setting do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 2: Predictors | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 2.1 Were predictors defined and assessed in a similar way for all participants? | | Yes |
| 2.2 Were predictor assessments made without knowledge of outcome data? | | Yes |
| 2.3 Are all predictors available at the time the model is intended to be used? | | Yes |
| **Risk of bias introduced by predictors or their assessment** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| Concern that the definition, assessment or timing of predictors in the model do not match the review question | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 3: Outcome | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 3.1 Was the outcome determined appropriately? | | Yes |
| 3.2 Was a pre-specified or standard outcome definition used? | | Yes |
| 3.3 Were predictors excluded from the outcome definition? | | Yes |
| 3.4 Was the outcome defined and determined in a similar way for all participants? | | Yes |
| 3.5 Was the outcome determined without knowledge of predictor information? | | Yes |
| 3.6 Was the time interval between predictor assessment and outcome determination appropriate? | | Yes |
| **Risk of bias introduced by the outcome or its determination** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the outcome, its definition, timing or determination do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| **DOMAIN 4: Analysis** | | |
|---|---|---|
| **Risk of Bias** | | |
| 4.1 Were there a reasonable number of participants with the outcome? | | Yes |
| 4.2 Were continuous and categorical predictors handled appropriately? | | Yes |
| 4.3 Were all enrolled participants included in the analysis? | | Yes |
| 4.4 Were participants with missing data handled appropriately? | | No |
| 4.5 Was selection of predictors based on univariable analysis avoided? | | Not applicable |
| 4.6 Were complexities in the data (e.g. censoring, competing risks, sampling of controls) accounted for appropriately? | | Yes |
| 4.7 Were relevant model performance measures evaluated appropriately? | | No information |
| 4.8 Were model overfitting and optimism in model performance accounted for? | | Not applicable |
| 4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis? | | Not applicable |
| **Risk of bias introduced by the analysis** | **RISK:** (low/ high/ unclear) | High |
| Rationale of bias rating:<br>Patients with missing data were excluded | | |

## Su et al.

| DOMAIN 1: Participants | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 1.1 Were appropriate data sources used, e.g. cohort, RCT or nested case-control study data? | | Yes |
| 1.2 Were all inclusions and exclusions of participants appropriate? | | Yes |
| **Risk of bias introduced by selection of participants** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the included participants and setting do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| DOMAIN 2: Predictors | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 2.1 Were predictors defined and assessed in a similar way for all participants? | | Yes |
| 2.2 Were predictor assessments made without knowledge of outcome data? | | Yes |
| 2.3 Are all predictors available at the time the model is intended to be used? | | Yes |
| **Risk of bias introduced by predictors or their assessment** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| Concern that the definition, assessment or timing of predictors in the model do not match the review question | **CONCERN:** (low/ high/ unclear) | High |
| Rationale of applicability rating:<br>The threshold value of NEWS was not consistent with other studies | | |

| DOMAIN 3: Outcome | | |
|---|---|---|
| **A. Risk of Bias** | | |
| 3.1 Was the outcome determined appropriately? | | Yes |
| 3.2 Was a pre-specified or standard outcome definition used? | | Yes |
| 3.3 Were predictors excluded from the outcome definition? | | Yes |
| 3.4 Was the outcome defined and determined in a similar way for all participants? | | Yes |
| 3.5 Was the outcome determined without knowledge of predictor information? | | Yes |
| 3.6 Was the time interval between predictor assessment and outcome determination appropriate? | | Yes |
| **Risk of bias introduced by the outcome or its determination** | **RISK:** (low/ high/ unclear) | Low |
| Rationale of bias rating:<br>Low risk of bias | | |
| **B. Applicability** | | |
| **Concern that the outcome, its definition, timing or determination do not match the review question** | **CONCERN:** (low/ high/ unclear) | Low |
| Rationale of applicability rating:<br>Match the review question | | |

| **DOMAIN 4: Analysis** | | |
|---|---|---|
| **Risk of Bias** | | |
| 4.1 Were there a reasonable number of participants with the outcome? | | Yes |
| 4.2 Were continuous and categorical predictors handled appropriately? | | Yes |
| 4.3 Were all enrolled participants included in the analysis? | | Yes |
| 4.4 Were participants with missing data handled appropriately? | | No information |
| 4.5 Was selection of predictors based on univariable analysis avoided? | | Not applicable |
| 4.6 Were complexities in the data (e.g. censoring, competing risks, sampling of controls) accounted for appropriately? | | Yes |
| 4.7 Were relevant model performance measures evaluated appropriately? | | No information |
| 4.8 Were model overfitting and optimism in model performance accounted for? | | Not applicable |
| 4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis? | | Not applicable |
| **Risk of bias introduced by the analysis** | **RISK:** (low/ high/ unclear) | Unclear |
| Rationale of bias rating:<br>Did not report the missing data and the calibration of NEWS | | |